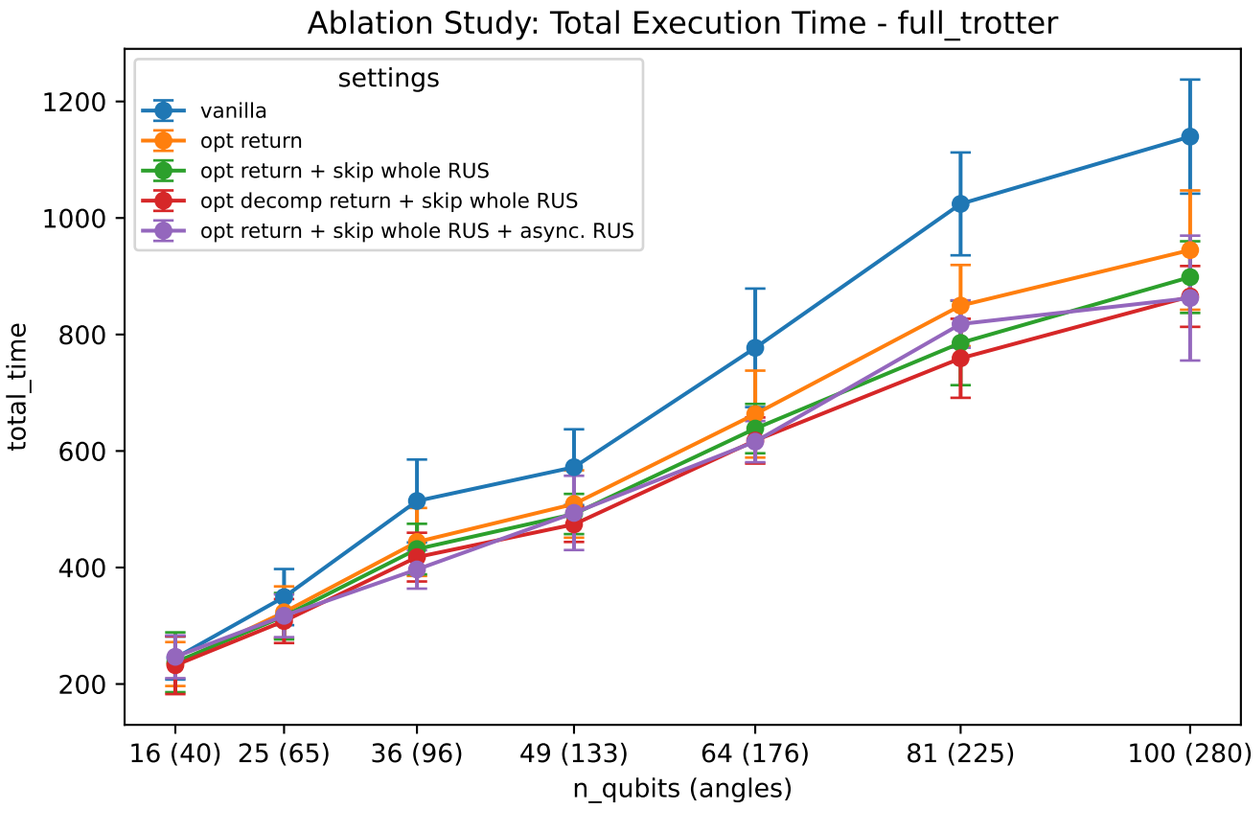
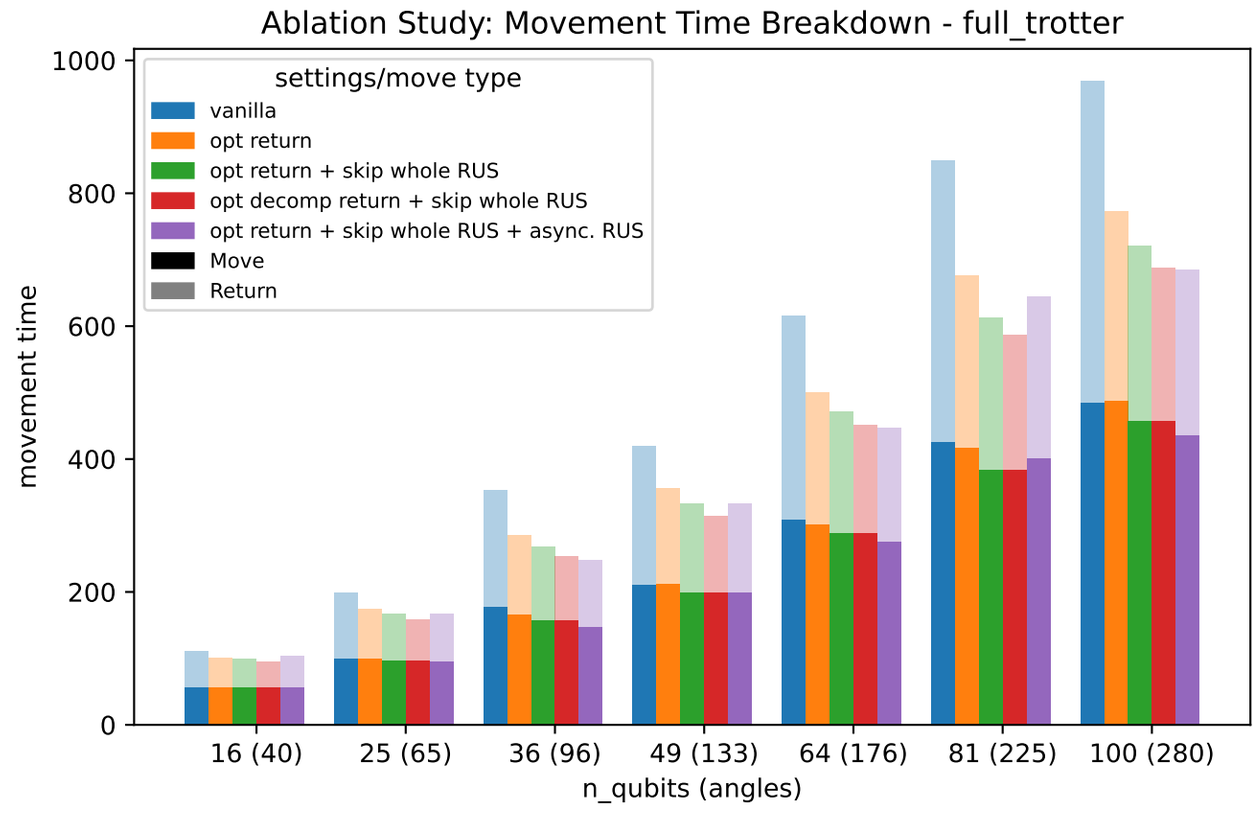
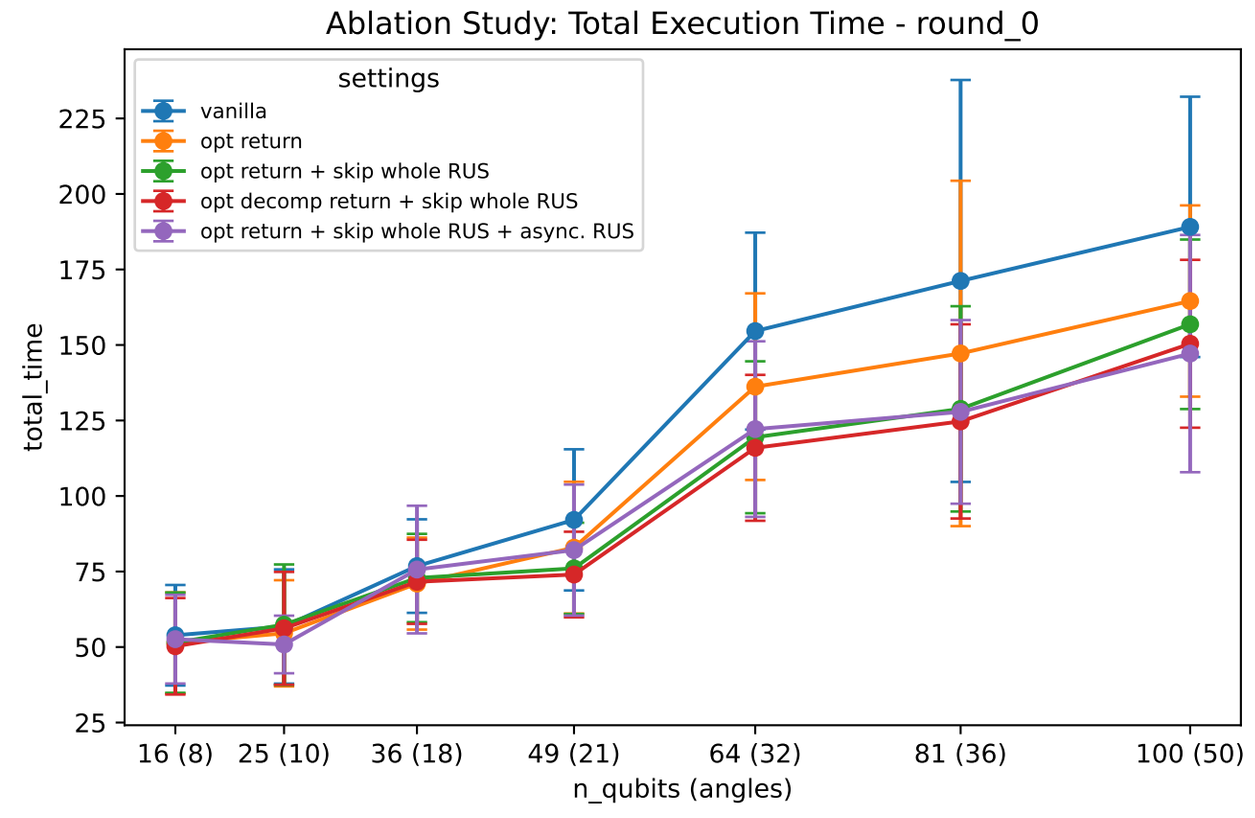
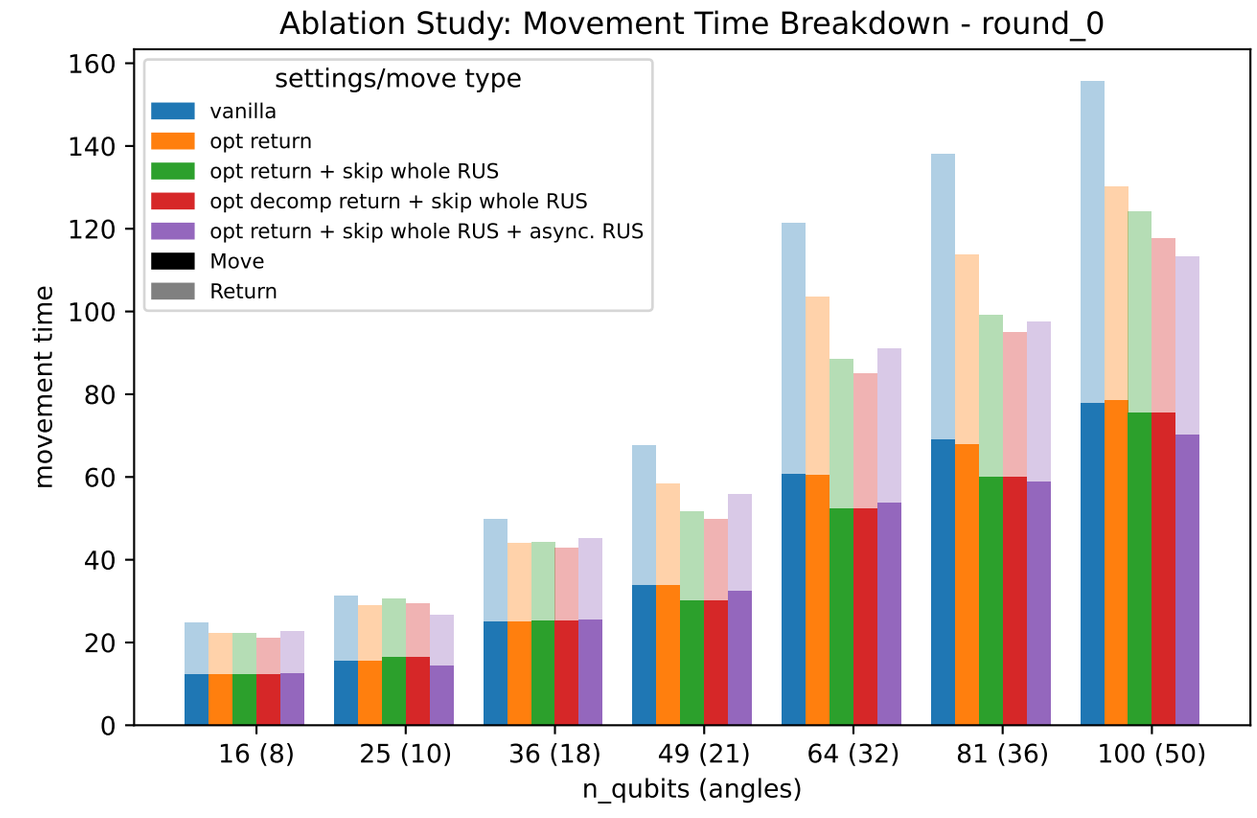
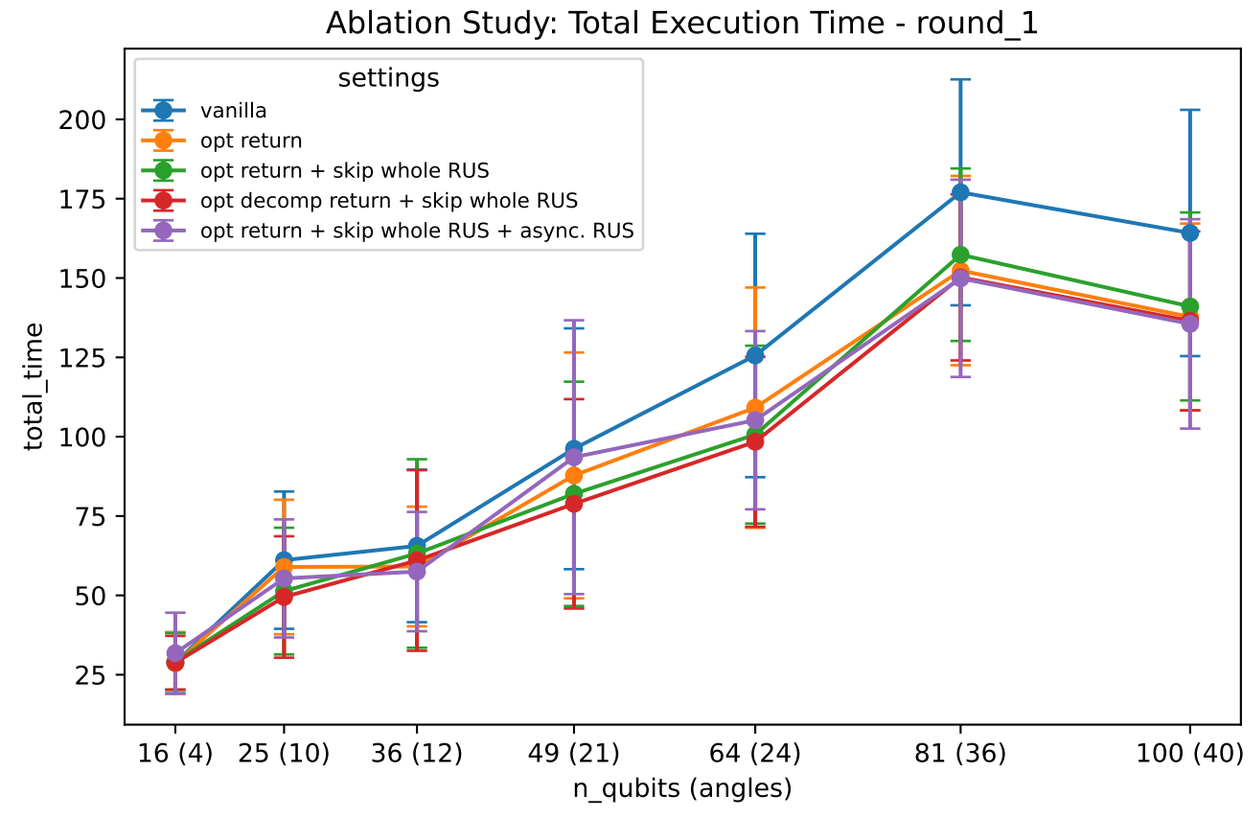
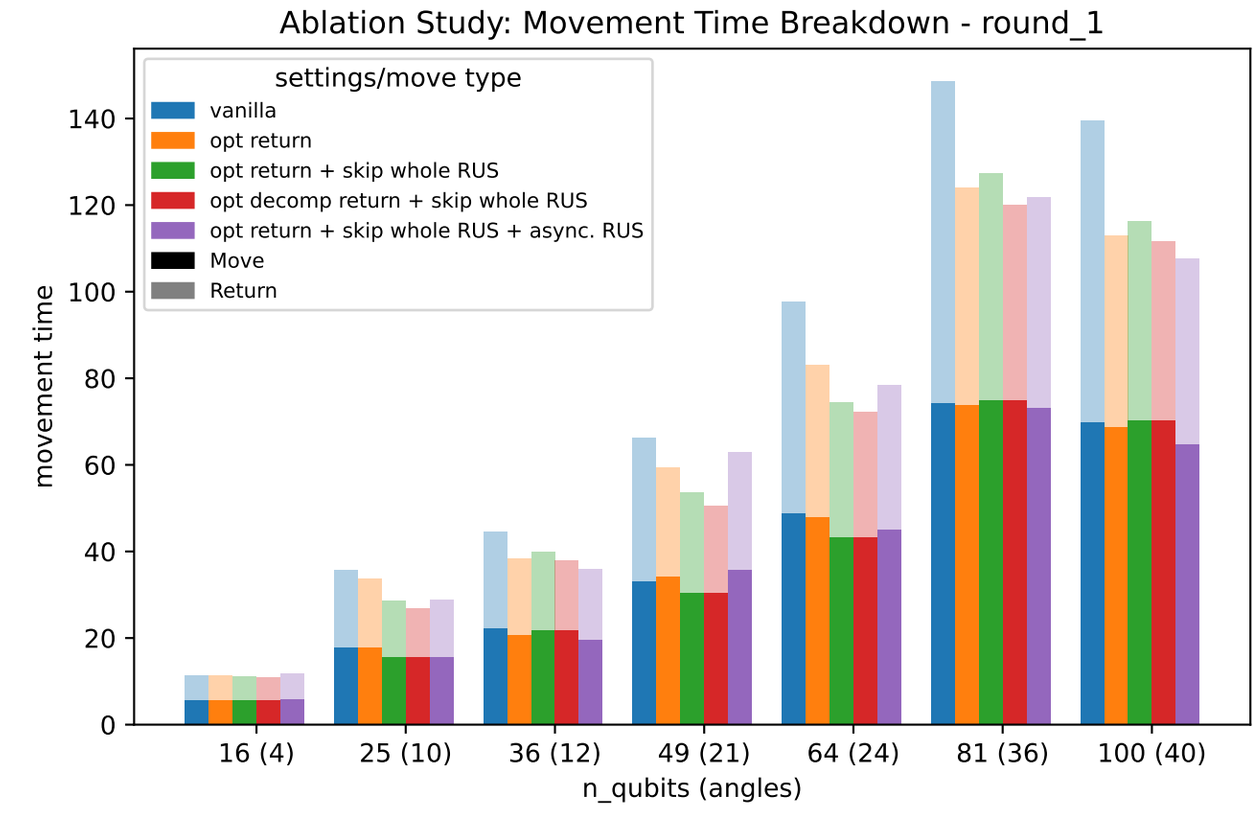

Ablation Study: Total Execution Time - full_trotter
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (40)
25 (65)
36 (96)
49 (133)
64 (176)
81 (225)
100 (280)

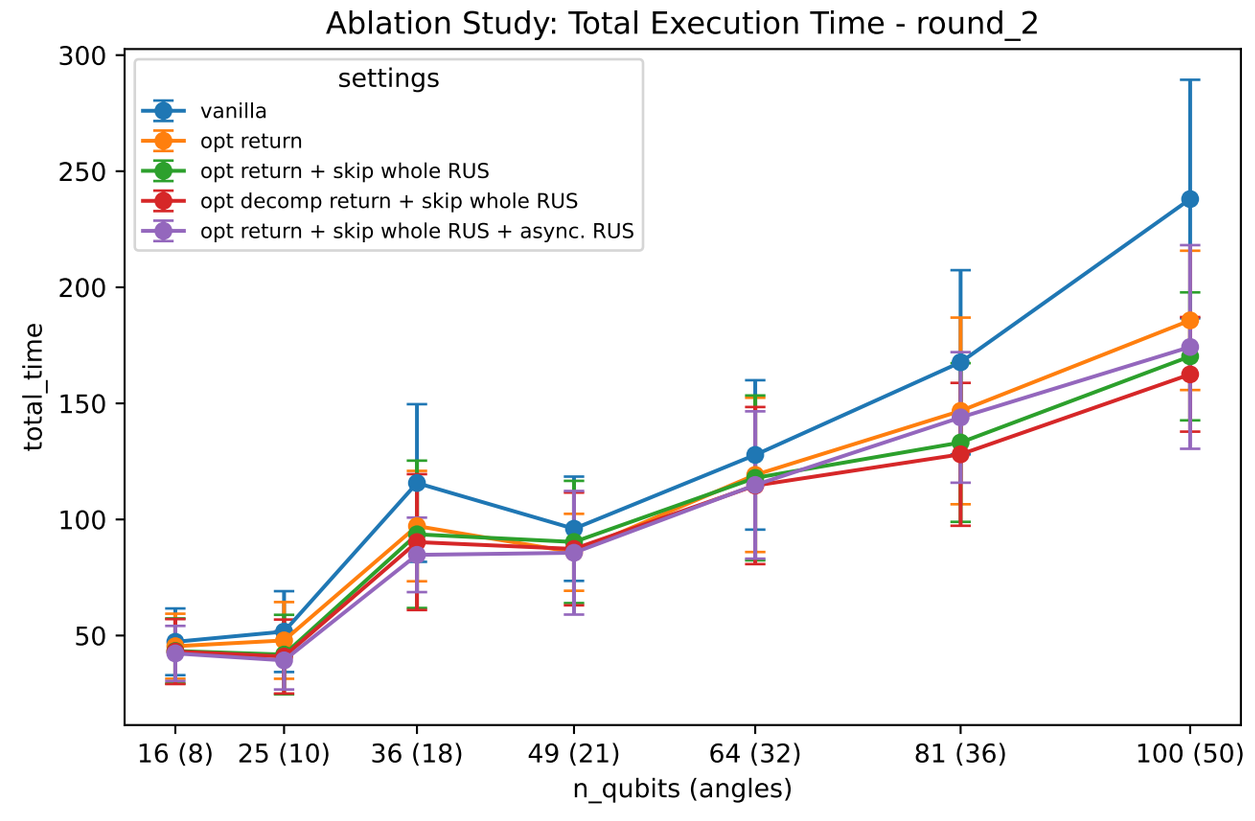

Ablation Study: Movement Time Breakdown - full_trotter
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (40)
25 (65)
36 (96)
49 (133)
64 (176)
81 (225)
100 (280)

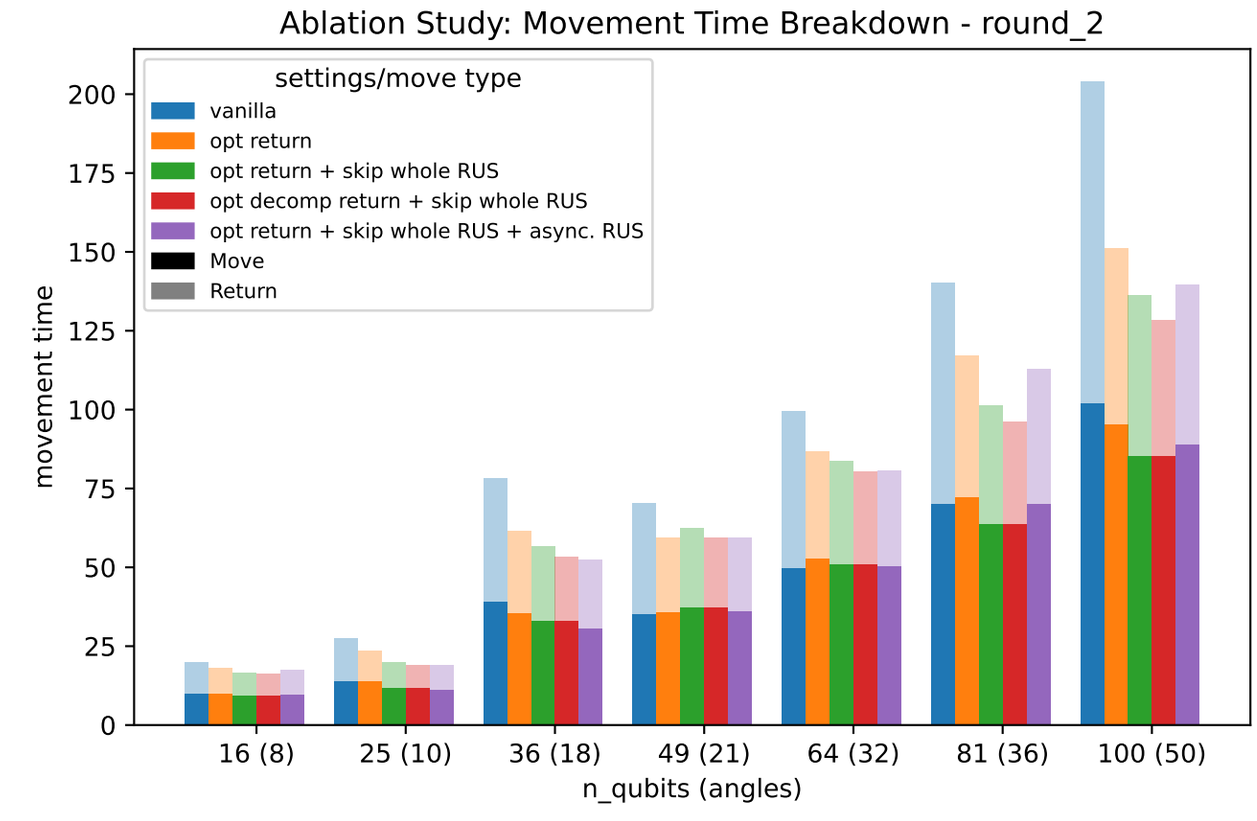

Ablation Study: Total Execution Time - round_0
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (8)
25 (10)
36 (18)
49 (21)
64 (32)
81 (36)
100 (50)

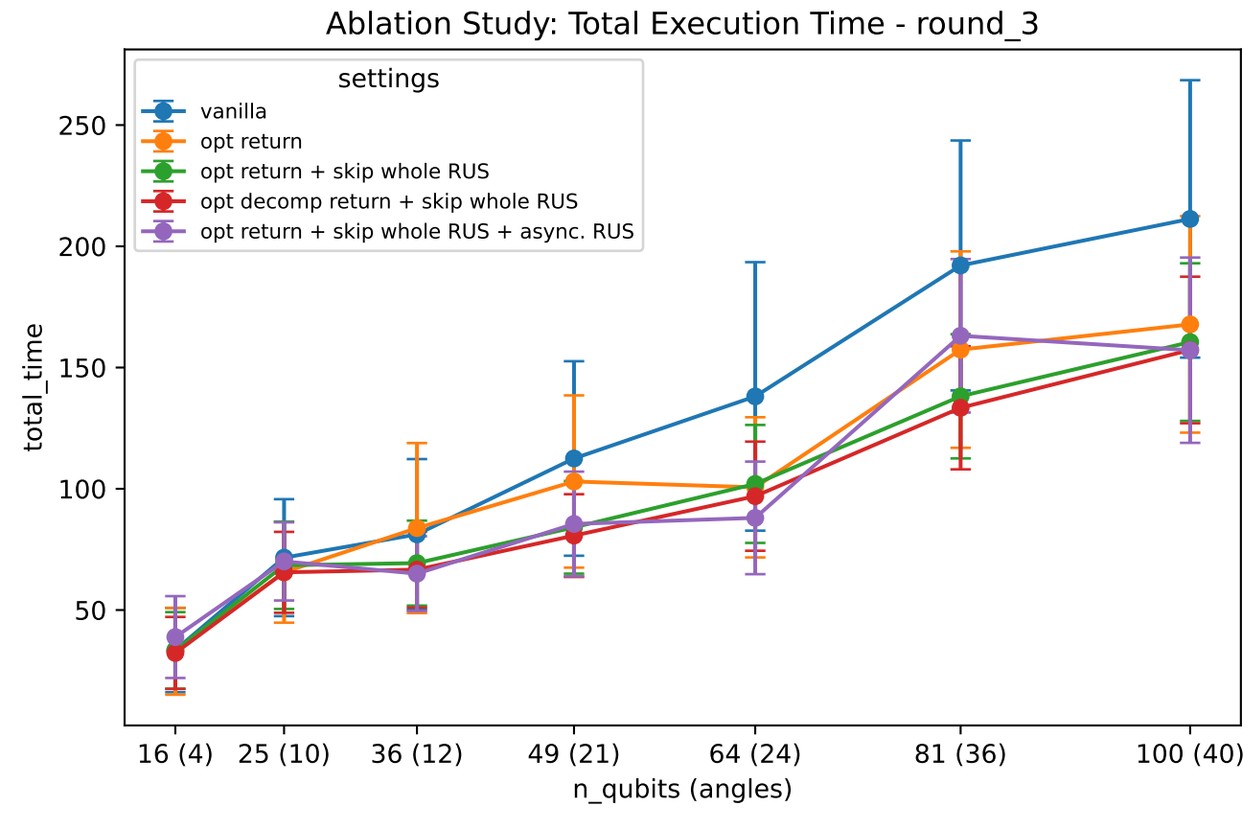

Ablation Study: Movement Time Breakdown - round_0
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (8)
25 (10)
36 (18)
49 (21)
64 (32)
81 (36)
100 (50)

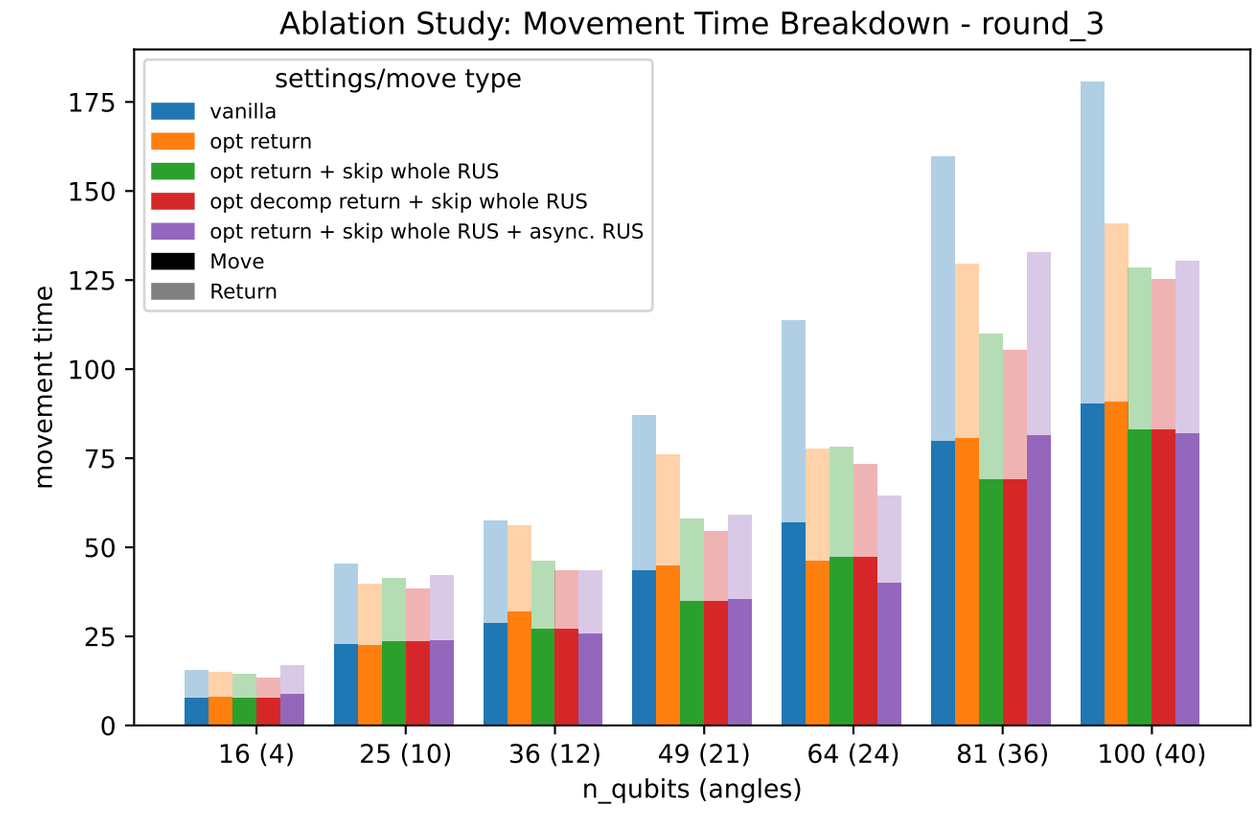

Ablation Study: Total Execution Time - round_1
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (4)
25 (10)
36 (12)
49 (21)
64 (24)
81 (36)
100 (40)

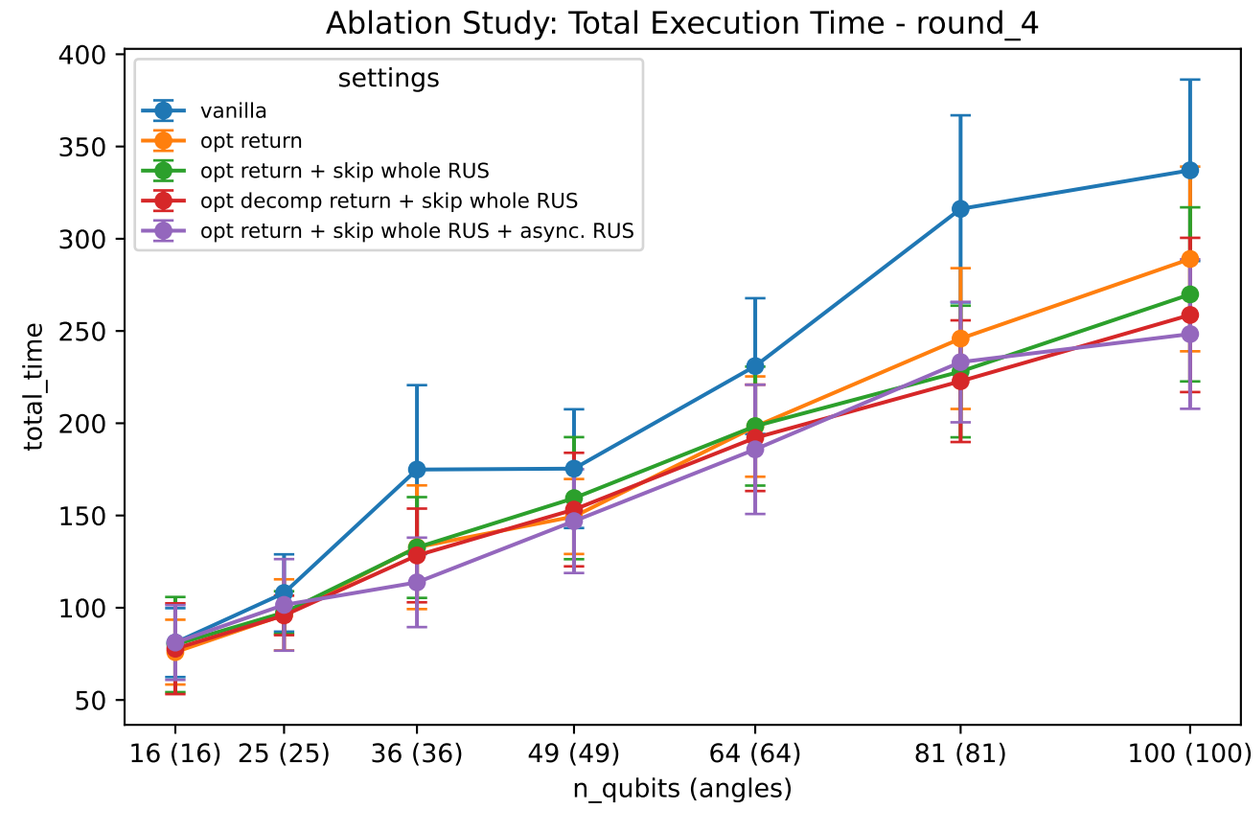

Ablation Study: Movement Time Breakdown - round_1
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (4)
25 (10)
36 (12)
49 (21)
64 (24)
81 (36)
100 (40)

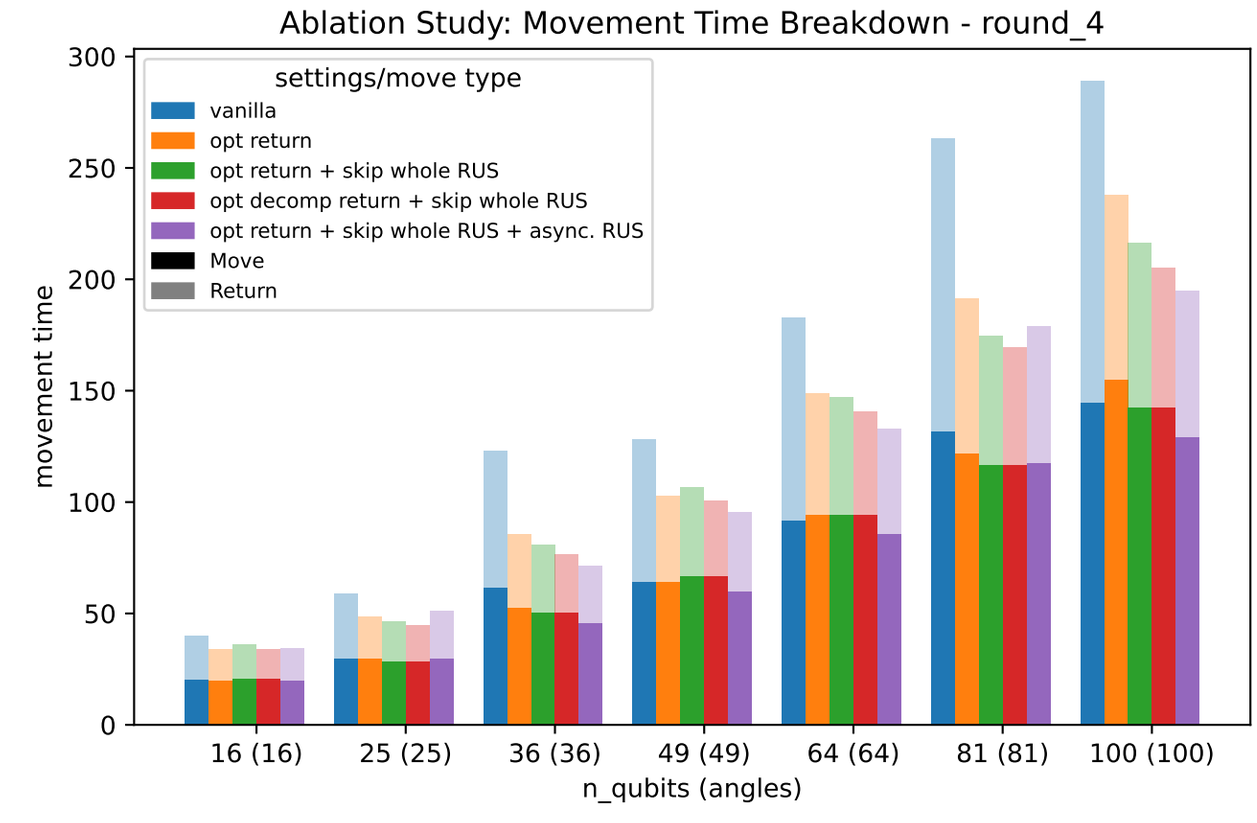

Ablation Study: Total Execution Time - round_2
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (8)
25 (10)
36 (18)
49 (21)
64 (32)
81 (36)
100 (50)


Ablation Study: Movement Time Breakdown - round_2
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (8)
25 (10)
36 (18)
49 (21)
64 (32)
81 (36)
100 (50)


Ablation Study: Total Execution Time - round_3
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (4)
25 (10)
36 (12)
49 (21)
64 (24)
81 (36)
100 (40)


Ablation Study: Movement Time Breakdown - round_3
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (4)
25 (10)
36 (12)
49 (21)
64 (24)
81 (36)
100 (40)


Ablation Study: Total Execution Time - round_4
settings
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
total_time
n_qubits (angles)
16 (16)
25 (25)
36 (36)
49 (49)
64 (64)
81 (81)
100 (100)


Ablation Study: Movement Time Breakdown - round_4
settings/move type
vanilla
opt return
opt return + skip whole RUS
opt decomp return + skip whole RUS
opt return + skip whole RUS + async. RUS
Move
Return
movement time
n_qubits (angles)
16 (16)
25 (25)
36 (36)
49 (49)
64 (64)
81 (81)
100 (100)